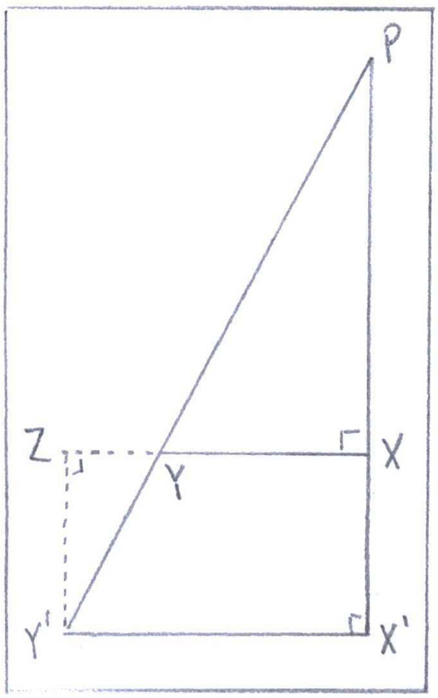
Chapter 5: History of Parallel Postulate:

Exercises:

1.

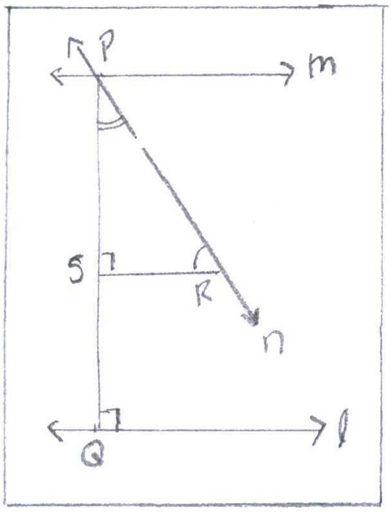


(pg 120) Obtuse Angle Hypothesis
"the angle sum of every triangle is greater than a 'straight' angle"

(from exercise) "the side opposite the acute angle at least doubled, whereas the side adjacent to the acute angle is at most doubled."

Proof:

Hypothesis	Evidence
1) A right triangle ΔPXY	RAA hypothesis
2) A point Z exists, so segment $\overline{XZ}$ has a midpoint at point Y.	(pg 12) "Definition - point"
3) $\overline{XY} \cong \overline{YZ}$	Proposition 4.3 (midpoints)
4) $\overline{PY'} = 2\overline{PY} = \overline{PY} + \overline{YY'}$	



"Euclid's Postulate V to Wallis' Postulate"

$\overline{PY} = \overline{YY'}$	
5) $\measuredangle PXY = \measuredangle YZY'$	Euclid's Postulate IV
6) $\Delta PXY \cong \Delta Y'ZY$	Axiom C-6 (S.A.S.)
7) The side opposite the acute angle doubled, $\overline{X'Y} \cong 2\overline{XY}$	Proposition 4.15
8) The side adjacent the acute angle doubled, $\overline{PX'} \cong 2\overline{PX}$ because Lamberts Quadrilateral $\square XX'Y'Z$	Corollary 3 (Saccher III, Corollary I) to Proposition 4.13

2. (from exercise) "Archimede's axiom implies Aristotle's axiom"

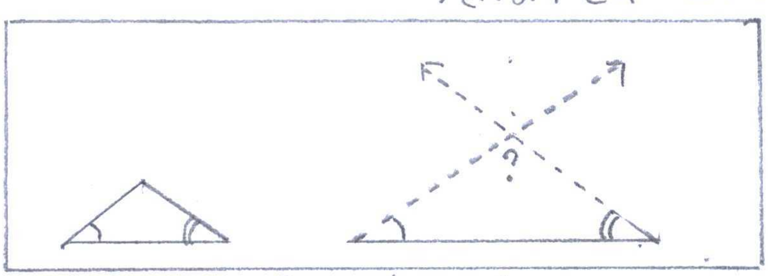
Proof:

Hypothesis	Evidence
1) A right triangle ΔPXY	RAA hypothesis
2) A point Z exists, so segment $\overline{XZ}$ has a midpoint at Y	Proposition 4.3 (midpoints)
3) $\overline{XY} \cong \overline{YZ}$	Euclid Postulate II
4) $\overline{PY'} = 2\overline{PY} = \overline{PY} + \overline{YY'}$, $\overline{PY} = \overline{YY'}$	Measurement Theorem 4.3

5) $\measuredangle PXY = \measuredangle YZY'$	Euclid's Postulate IV
6) $\Delta PXY \cong \Delta Y'ZY$	Axiom C-6 (S.A.S)
7) The side opposite the acute angle doubled, $\overline{X'Y} \cong 2\overline{XY}$	Proposition 4.13
8) The side adjacent the acute angle doubled, $\overline{PX'} \cong 2\overline{PX}$ because Lamberts Quadrilateral $\square XX'Y'Z$	Corollary 3 (Saccheri III, Corollary I) to Proposition 4.13
9) A third triangle has triple, quadruple, or n-length hypotenuse ΔPY^nX^n	Archimedes Axiom
10) $\overline{PX}^n \geq 2^n\overline{PX}$ and $\overline{PY}^n \geq 2^n\overline{PY}$	Aristotle's Axiom
11) $\overline{PY}^n > \overline{PX}^n$	Aristotle's Axiom

3.

a) (from exercise) "Euclids fifth postulate implies Wallis' postulate."



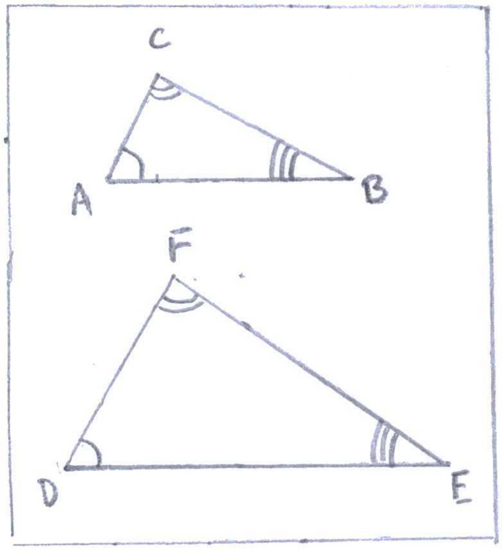
"Wallis' Postulate"

Proof Hypothesis	Evidence
1) $\overleftrightarrow{m} \parallel \overleftrightarrow{k}$	Euclid's Postulate V
2) Point P is on line m	(pg) "Definition - Point"
3) PQ is the trav-ersal to lines $\overleftrightarrow{m}$ and $\overleftrightarrow{l}$.	Proposition 4.9
4) Line $\overleftrightarrow{n}$ is the other line through point P	Proposition 2.1
5) On line $\overleftrightarrow{n}$ is a point R, for a right triangle ΔPSR along PQ.	(pg 98) "Definition - Triangle
6) $\overleftrightarrow{n}$ meets $\overleftrightarrow{l}$ at point T	RAA hypothesis
7) $\angle TPQ \cong \angle RPS$	Axiom C-4
8) $\overline{PQ} \cong \overline{PS}$	Axiom C-1
9) $\overline{PR} \cong \overline{PT}$	Axiom C-1
10) $\Delta PSR \sim \Delta PQT$	RAA conclusion Wallis' Postulate

(pg 215) "Definition-
similar"

b) (From exercise) Wallis' Postulate - replaced

"Given any triangle ΔABC and given any segment $AB \cong DE$, there exists a triangle ΔDEF having DE as one of its sides such $\Delta ABC \cong \Delta DEF$."

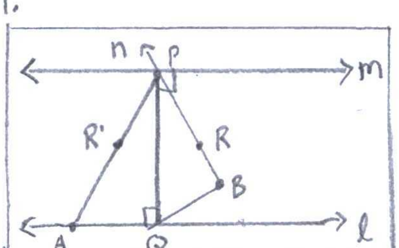


"Similar Triangles"

Proof:

Hypothesis	Evidence
1) Segments $\overline{AB}$ and DE form ΔABC and ΔDEF	(pg 98) "Definition-Triangle"
2) $\measuredangle CAB \cong \measuredangle FDE$	Axiom C-4
3) $\measuredangle ACB \cong \measuredangle DEF$	Proposition 3.17 (A.S.A. criterion for congruence).
4) $\Delta ABC \cong \Delta DEF$	(pg 119) "Definition-congruence"

4.



"Legendre's Parralel"

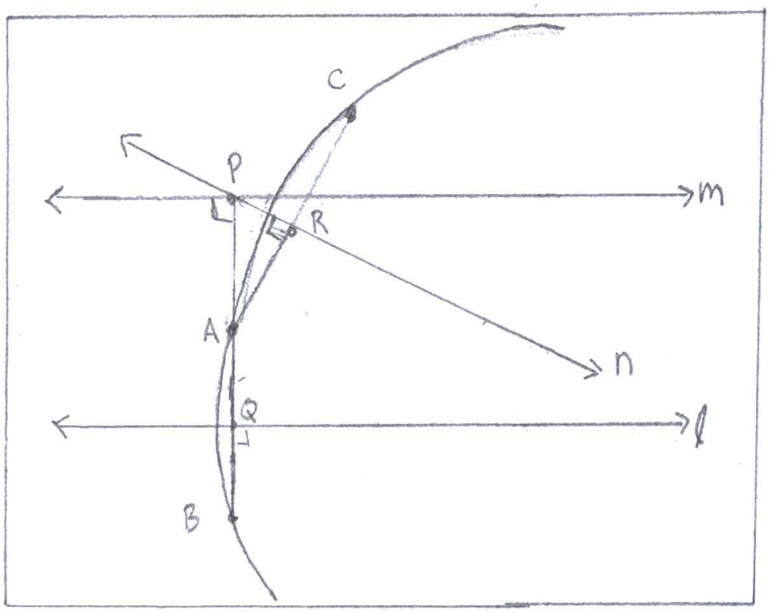
(pg 52) Legendres Parallel Postulate

Flaw #1: Side-angle-side later acknowledged by Axiom C-6 (SAS).

Flaw #2: Perpendicular requires right angles on page 42.

5.

Flaw #3: Interior of an angle defined not by rays, but points (page 162).



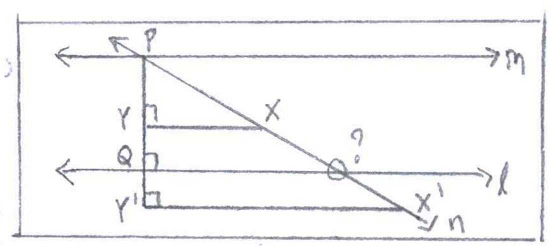
"Farkas Bolyai"

(weak support)

Farkas had a bad argument unless $\overline{AR} = \overline{AQ}$. The circle looks real, but not exact by compass and ruler.

6.

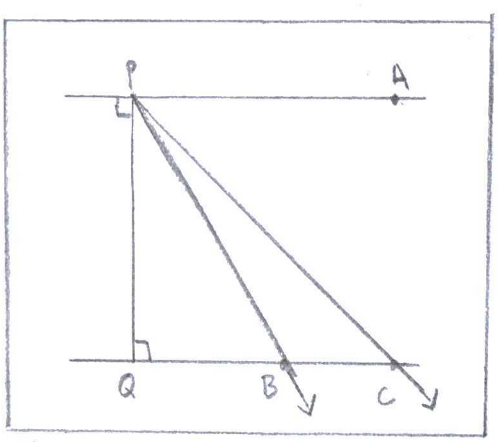
In hyperbolic geometry, ray PX never crosses line $\overleftrightarrow{l}$. Where limiting ray?



"Proclus"

7.

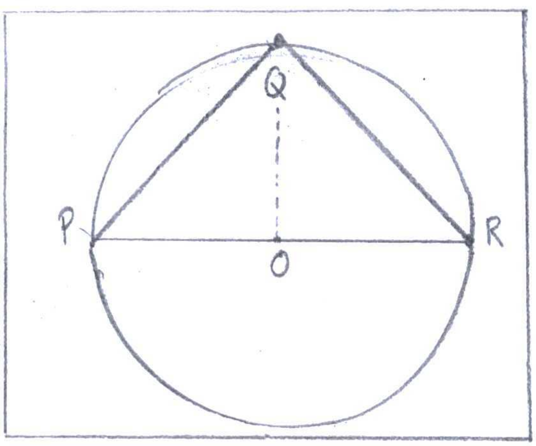
J.D. Gergonne assumed at the end, "Hence all rays between $\overrightarrow{PA}$ and $\overrightarrow{PQ}$ meet l." Where limiting ray?



"J.D. Gergonne"

8. Proclus' problem was collinear. Circumstances lend Y on opposite side to l. Within the conditions, $Y * Z * X$ wasn't equivalent to $X * Y * Z$. Secondly, Y could be opposite to m across l.

9. (from exercise) Existence of a Triangle whose angle sum is 180°



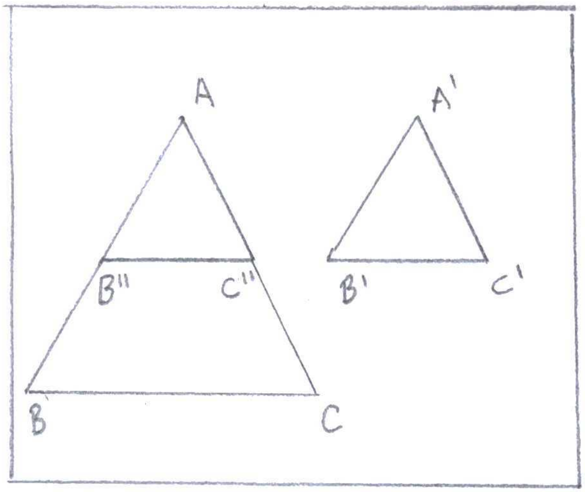
"Euclidean Geometry"

"If points P, Q, R lie on a circle with center O, and if $\sphericalangle PQR$ is acute, then $(\sphericalangle PQR)^\circ = \frac{1}{2}(\sphericalangle POR)^\circ$."

Proof: Hypothesis	Evidence
1) A circle with radius $\overline{OP} = \overline{OQ} = \overline{OR}$	Euclid Postulate III
2) $P * O * R$ is 180°	(pg 19) "Definition - Supplementary"
3) $\sphericalangle PQR = \frac{1}{2}(\sphericalangle POR)$	Euclid's Postulate III.20

10.

(from exercise) Fundamental theorem on Similar Triangles



"Similar triangles"

"Given $\Delta ABC \sim \Delta A'B'C'$, i.e., given $\measuredangle A \cong \measuredangle A'$, $\measuredangle B \cong \measuredangle B'$, and $\measuredangle C \cong \measuredangle C'$. Then corresponding sides are proportional; i.e. $AB/A'B' = AC/A'C' = BC/B'C'$"

Proof: Hypothesis	Evidence
1) From information given	RAA hypothesis
2) A point B'' is between A and B, so $\overline{AB''} \cong \overline{A'B'}$	Euclid Postulate II
3) A point C'' is between A and C, so $\overline{AC''} \cong \overline{A'C'}$	Euclid Postulate II
4) $\Delta AB''C'' \cong \Delta A'B'C'$	Axiom C-6 (SAS)
5) $\overline{B''C''} \cong \overrightarrow{B'C'}$	(pg 193) "Definition - congruent"
6) $\frac{\overline{AB''}}{\overline{B''B}} = \frac{\overline{A'B'}}{\overline{B''B}}$, $\frac{\overline{AC''}}{\overline{C''C}} = \frac{\overline{A'C'}}{\overline{C''C}}$ and $\frac{\overline{BC}}{\overline{B''C''}} = \frac{\overline{B'C'}}{\overline{B''C''}}$	(pg 232) "Parallel Projection Theorem"

7) $\frac{\overline{AB''}}{\overline{A'B'}} = \frac{\overline{AC''}}{\overline{A'C'}} = \frac{\overline{BC}}{\overline{B'C'}}$ which equates to $\frac{\overline{AB}}{\overline{A'B'}} = \frac{\overline{AC}}{\overline{A'C'}} = \frac{\overline{BC}}{\overline{B'C'}}$	RAA result

11. (from exercise) Converse to Fundamental Theorem on Similar Triangles

"Corresponding angles are proportional, so $\Delta ABC \sim \Delta A'B'C'$."

Proof Hypothesis	Evidence
1) By the supplement given $AB/A'B' = \ldots$ $\ldots \overline{AC}/\overline{A'C'} = \overline{BC}/\overline{B'C'}$ for ΔABC and $\Delta A'B'C'$	RAA hypothesis
2) Point B'' is between A and B, so $AB'' \cong A'B'$	Euclids Postulate II
3) Point C'' is between A and C, so $AC'' \cong A'C'$	Euclids Postulate II
4) $\overleftrightarrow{B''C''}$ passes through AB and AC parallel to $\overleftrightarrow{BC}$	Pasch's Theorem

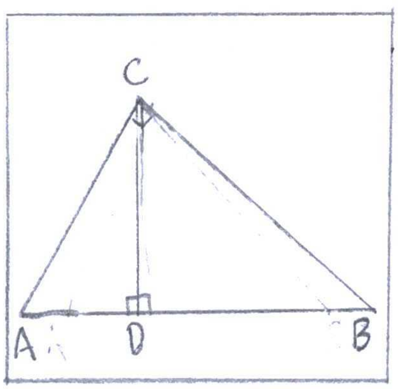
5) $\triangle AB''C \cong \triangle A'B'C'$	Proposition 3.22 (S.S.S. criterion for congruence)
6) $\triangle ABC \sim \triangle A'B'C'$	Parallel Projection Theorem, RAA result

12. (from exercise) S.A.S. similarity Criterion
"If $\angle A \cong \angle A'$ and $AB/A'B' = AC/A'C'$, then $\triangle ABC \sim \triangle A'B'C'$"

Proof: Hypothesis	Evidence
1) Two triangles materialize with $\angle A \cong \angle A'$ as $AB/A'B' = AC/A'C'$ in $\triangle ABC$ and $\triangle A'B'C'$	RAA hypothesis
2) Point B'' is between A and B, so $AB'' \cong A'B'$	Euclids Postulate II
3) Point C'' is between A and C, so $AC'' \cong A'C'$	Euclids Postulate II
4) $\triangle AB''C'' \cong \triangle A'B'C'$	Axiom C-6 (S.A.S)

5) $B''C'' \parallel BC$	Euclid's Postulate V
6) $\frac{AB''}{B''B} = \frac{A'B'}{B''B}$, $\frac{AC''}{C''C} = \frac{A'C'}{C''C}$, and $\frac{BC}{B''B} = \frac{B'C'}{B''B}$	"Parallel Projection Theorem"
7) $\triangle ABC \sim \triangle A'B'C'$	RAA result

13.



"Pythagorean Law"

(from exercise) Pythagorean Law

$$(AB)^2 = (AC)^2 + (BC)^2$$

Proof — Hypothesis	Evidence
1) $\triangle ABC$ is a right triangle	(pg 19) "Definition - right triangle"
2) CD is the alititude to the hypotenuse by a perpendicular segment.	(pg 25) "Definition - alititude"
3) $\measuredangle ADC + \measuredangle BDC = 180°$, so $\measuredangle ADC = \measuredangle BDC = 90°$	(pg 19) "Definition - Supplementary"
4) $\measuredangle A + \measuredangle ADC + \measuredangle DCA = 180°$	Proposition 4.11

5) $\measuredangle B + \measuredangle BDC + \measuredangle DCB = 180^\circ$	Proposition 4.11
6) $\measuredangle A + \measuredangle B + \measuredangle C = 180^\circ$	Proposition 4.11
7) $\triangle ACD \sim \triangle ABC \sim \triangle CBD$	(pg 215) "Definition-similar"
8) $(AB)^2 = AB \cdot (AD + DB) = (AC)^2 + (BC)^2$	RAA outcome

14. (from exercise) Law of Sines

$$\frac{a}{b} = \frac{\sin(\measuredangle A)}{\sin(\measuredangle B)}$$

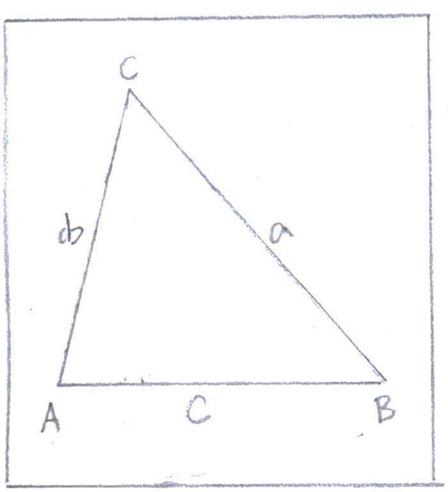
Proof:

$$\sin(\measuredangle A) = \frac{BC}{AB} = \frac{a}{AB}$$

$$\sin(\measuredangle B) = \frac{AC}{AB} = \frac{b}{AB}$$

$$\frac{\sin(\measuredangle A)}{\sin(\measuredangle B)} = \frac{\left(\frac{a}{AB}\right)}{\left(\frac{b}{AB}\right)}$$

$$= \frac{a}{b}$$



"Law of sines and cosines"

(From exercise) "Law of Cosines

$$c^2 = a^2 + b^2 - 2ab\cos(\measuredangle C)$$

Proof: $a^2 + b^2 - 2ab\cos(\measuredangle C) = (AB\sin(\measuredangle A))^2 + (AB\sin(\measuredangle B))^2$

$$- 2(AB)^2\sin(\measuredangle A)\sin(\measuredangle B)\cos(90^\circ)$$

$$= (AB)^2\sin^2(\measuredangle A) + (AB)^2\sin^2(\measuredangle B)$$

$$= c^2[\sin^2(\measuredangle A) + \sin^2(\measuredangle B)]$$

$$= c^2$$

(From exercise) Converse to Pythagoreans Theorem

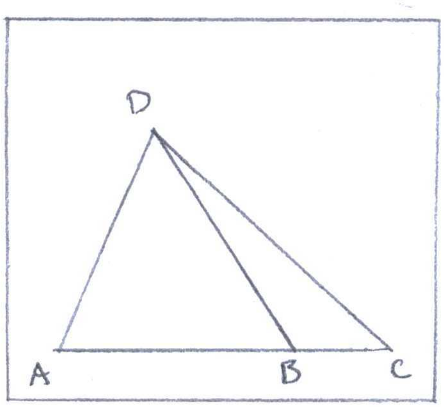
$$a^2 + b^2 = c^2$$

Proof: $a^2 + b^2 = (AB)^2\sin^2(\measuredangle A) + (AB)^2\sin^2(\measuredangle B)$

$$= c^2[\sin^2(\measuredangle A) + \sin^2(\measuredangle B)]$$

$$= c^2$$

15.



"Law of Sines"

AB/BC by relating triangles:

$$\frac{AB}{DB} = \frac{\sin(\measuredangle ADB)}{\sin(\measuredangle BAD)} \quad || \quad \frac{BC}{DB} = \frac{\sin(\measuredangle CDB)}{\sin(\measuredangle BCD)}$$

$$AB = DB\frac{\sin(\measuredangle ADB)}{\sin(\measuredangle BAD)} \qquad DB = BC\frac{\sin(\measuredangle BCD)}{\sin(\measuredangle CDB)}$$

$$\frac{AB}{BC} = \frac{AD\sin(\measuredangle ADB)}{CD\sin(\measuredangle CDB)}$$

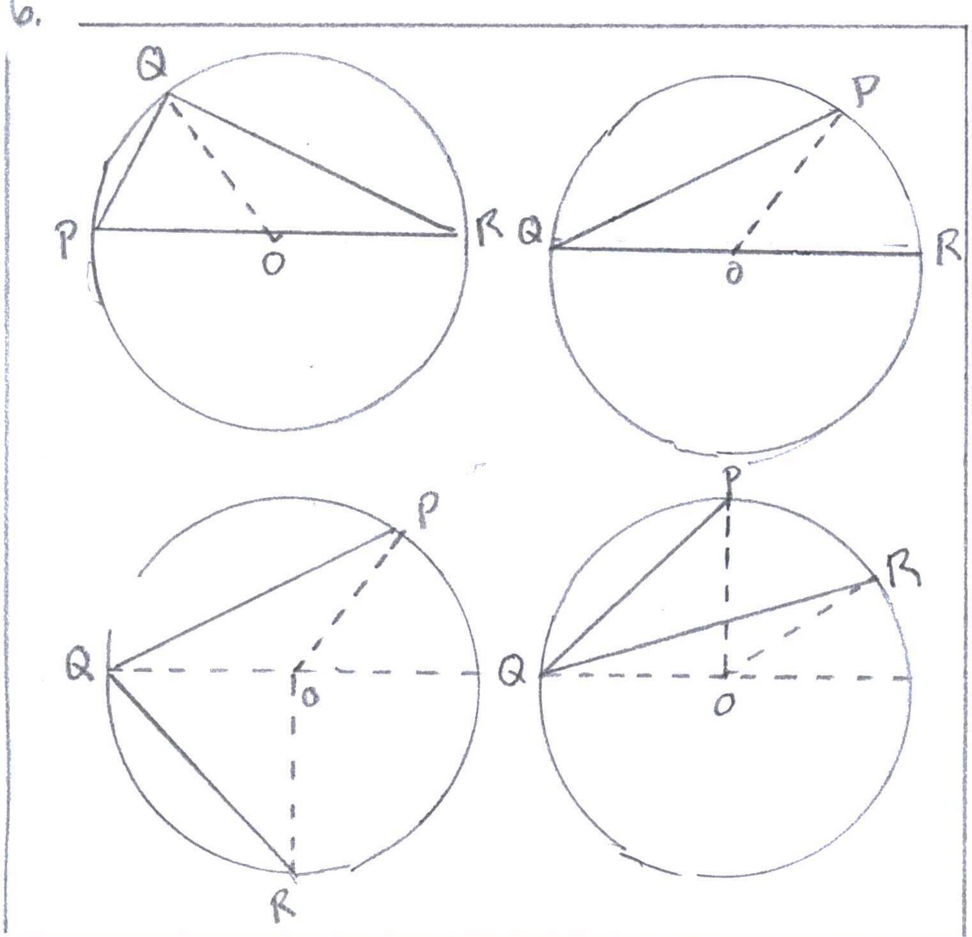
AC/BC by relating triangles:

$$\frac{AC}{DC} = \frac{\sin(\measuredangle ADC)}{\sin(\measuredangle DAC)} \quad \Bigg\| \quad \frac{BC}{DC} = \frac{\sin(\measuredangle BDC)}{\sin(\measuredangle CBD)}$$

$$AC = DC\,\frac{\sin(\measuredangle ADC)}{\sin(\measuredangle DAC)} \qquad DC = BC\,\frac{\sin(\measuredangle CBD)}{\sin(\measuredangle BDC)}$$

$$\frac{AC}{BC} = \frac{AD\sin(\measuredangle ADC)}{BD\sin(\measuredangle BDC)}$$

16.



(from exercise)

"If P and R are diametrically opposite, the $\measuredangle PQR$ is a right angle, and if O and Q are on the same side of PR, the $(\measuredangle PQR)^\circ = \frac{1}{2}(\measuredangle POR)^\circ$

$$(\measuredangle PQR)^\circ = \frac{1}{2}(\measuredangle POR)^\circ$$

Proof:

Hypothesis	Evidence
Case 1: $P * O * R$	
1) $P * O * R$ is 180°	(pg 19) "Definition - Supplementary"
2) $\measuredangle PQR = \frac{1}{2}(\measuredangle POR)$	Euclid III.20

$= 90°$

Case 2: $Q * O * R$

1) $(\measuredangle PQR)° = \frac{1}{2}(\measuredangle POR)°$ | Euclid III.20

Case 3: R and P are opposite sides to QO

1) $(\measuredangle PQR)°$ | Euclid III.20

$= (\measuredangle OQP + \measuredangle OQR)$

$= \frac{1}{2}(180° - \measuredangle QOR)$

$+ \frac{1}{2}(180° - \measuredangle POQ)$

$= \frac{1}{2}(\measuredangle PQR$

Case 3: P and R are on the same side QO

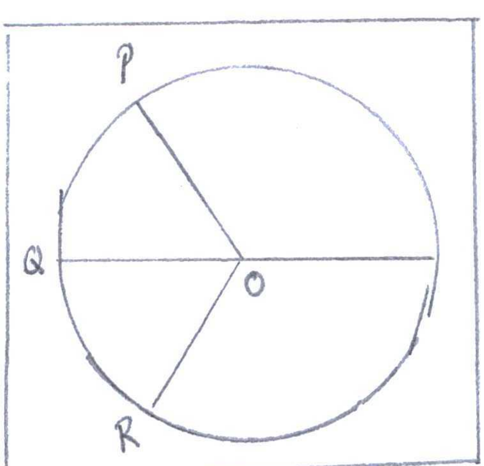
1) $(\measuredangle PQR)°$ | Euclid III.20

$= (\measuredangle PQO - \measuredangle OQR)$

$= \frac{1}{2}(180° - \measuredangle POQ)$

$- \frac{1}{2}(180° - \measuredangle QOR)$

$= \frac{1}{2}(\measuredangle POR)$



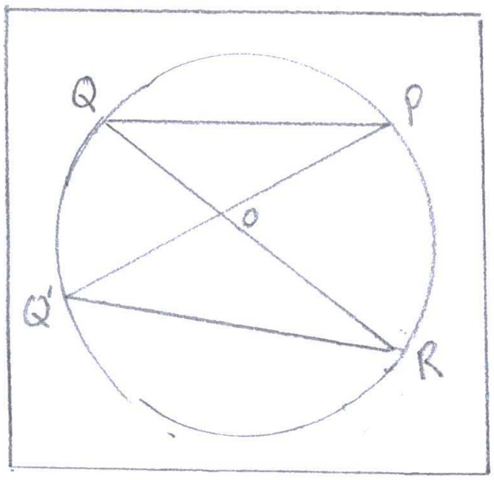
(from exercise)

"O and Q are on opposite sides of PR, so $(\measuredangle PQR)° = \frac{1}{2}(180° - \measuredangle POR)°$"

Proof:

Hypothesis	Evidence
1) $(\measuredangle PQR)^\circ$: $= \measuredangle PQO + \measuredangle OQR$ $= \frac{1}{2}(180^\circ - \measuredangle POQ) + \frac{1}{2}(180^\circ - \measuredangle QOR)$ $= \frac{1}{2}(180^\circ - \measuredangle POR)$	Euclid III.20

17.



$\measuredangle PQR \cong \measuredangle PQ'R$

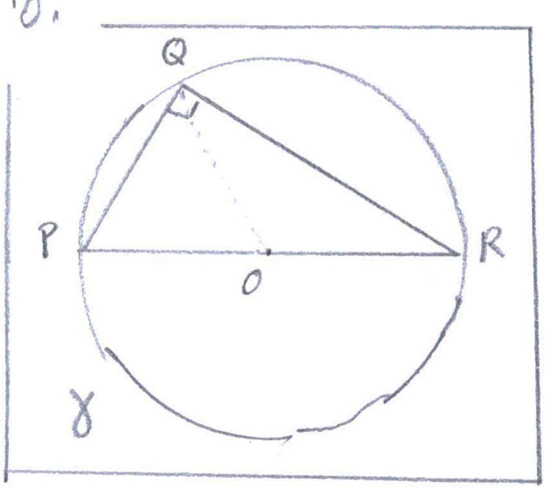
(from exercise)

"If two angles inscribed in a circle subtend the same arc, then they are congruent."

Proof:

Hypothesis	Evidence
1) By the powers that be	RAA hypothesis
2) $\measuredangle POQ = \measuredangle ROQ'$	(pg 31) Euclid I.15 (vertical angles)
3) $\measuredangle PQR = \frac{1}{2}(\measuredangle POR)$ $= \measuredangle PQ'R$	Euclid III.20

18.



(from exercise)

"If $\measuredangle PQR$ is a right triangle, then Q lies on the circle γ having PR as the diameter."

Proof

Hypothesis	Evidence
1) $\measuredangle POR = \measuredangle POQ + \measuredangle QOR = 180^\circ$	(pg 19) "Definition - Supplementary"
2) $\measuredangle PQR = \frac{1}{2}(\measuredangle POR) = 90^\circ$	Euclid III.20